3. Write program demonstrates how to perform morphological analysis using the NLTK library in Python.

Aim: To perform morphological analysis by converting words into their dictionary root forms using NLTK.

Procedure:

1. Install NLTK using python -m pip install nltk.
2. Import the WordNet lemmatizer.
3. Download the WordNet dataset.
4. Provide words with their parts of speech.
5. Display the root forms.

Code:

```
import nltk

nltk.download('wordnet', quiet=True)

from nltk.stem import WordNetLemmatizer

lem = WordNetLemmatizer()

words = ["cars", "playing", "studies", "ate"]

print("Original Word - Root Word")

print("cars -", lem.lemmatize("cars", pos="n"))

print("playing -", lem.lemmatize("playing", pos="v"))

print("studies -", lem.lemmatize("studies", pos="v"))

print("ate -", lem.lemmatize("ate", pos="v"))
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/3.py"
Original Word - Root Word
cars - car
playing - play
studies - study
ate - eat
```

Result: Morphological analysis was performed successfully using NLTK lemmatization.